

Systems-Level Reliability and Robustness Evaluation Framework for Document AI

Master's Thesis Defense

Tifang Desmond Ngoe

Czestochowa University of Technology

Supervisor: Prof. Piotr Duda

Academic Year 2025-2026

The Document AI Challenge

Enterprise Bottleneck: Modern pipelines rely heavily on extracting unstructured data from complex PDFs.

Visual Complexity: Dense tabular grids and multi-column geometries defy simple text reading orders.

The Perception Gap: Semantic LLMs without spatial awareness hallucinate structural data.

Critical Risk: A minor layout misread (e.g., distorting \$1,240.50) triggers catastrophic failures in downstream financial audits.

Data as the Core Foundation of AI

The Ultimate Catalyst: All downstream cognitive reasoning is fundamentally capped by the accuracy of the perception layer.

Bridging the Disconnect: We must bridge the gap where LLMs show advanced linguistic reasoning but fail at structural, spatial perception.

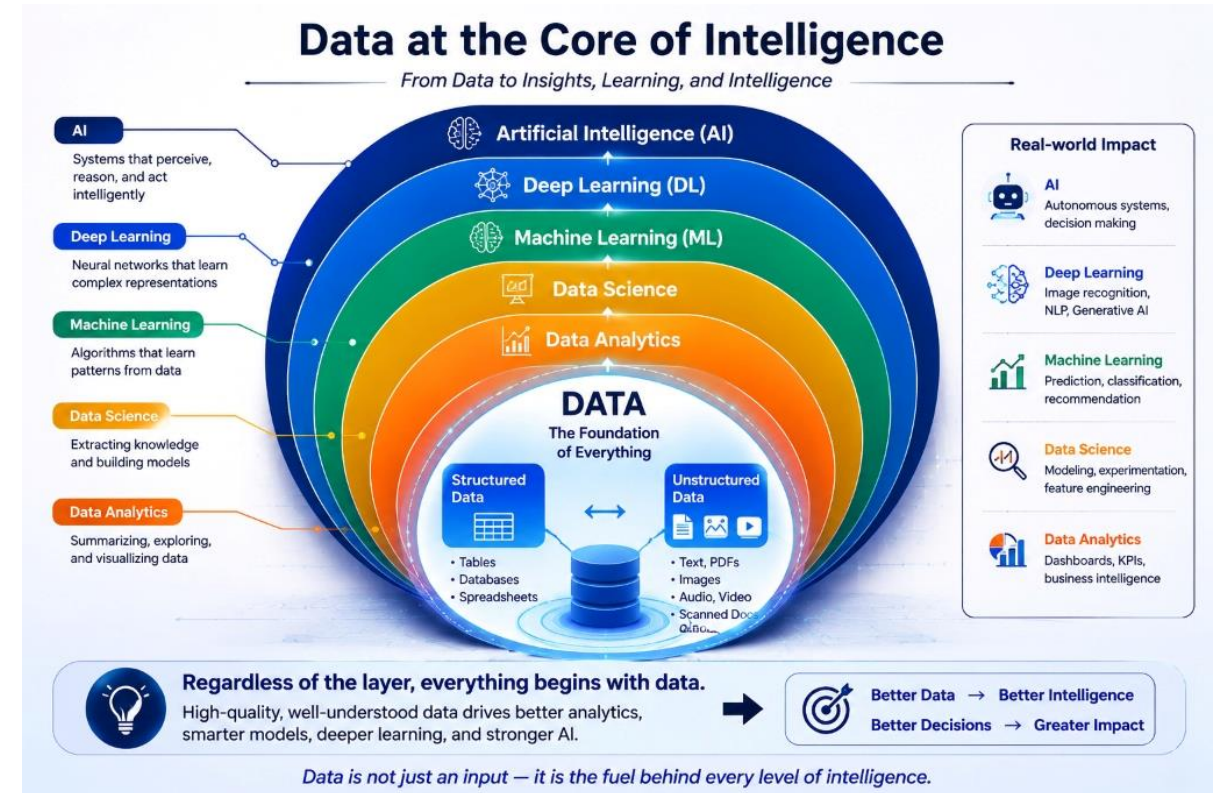


Figure 1.0: Data as the Core Foundation of AI Systems

Research Objective

Formulate a Benchmark: Create a rigorous reliability evaluation framework for DocVQA architectures under zero-shot conditions.

Investigate Perception: Isolate and evaluate the exact impact of spatial grounding on LLM hallucination rates.

Establish a Frontier: Map the accuracy-efficiency trade-offs across Traditional OCR, VLMs, and Hybrid models.

Traditional Baseline: OCR (Tesseract)

Heuristic Algorithms: Relies on rigid rules to group characters.

Linearization Failure: Converts multi-column layouts into a single, corrupted horizontal text stream.

Fidelity Loss: Completely destroys spatial relationships inside complex tables.

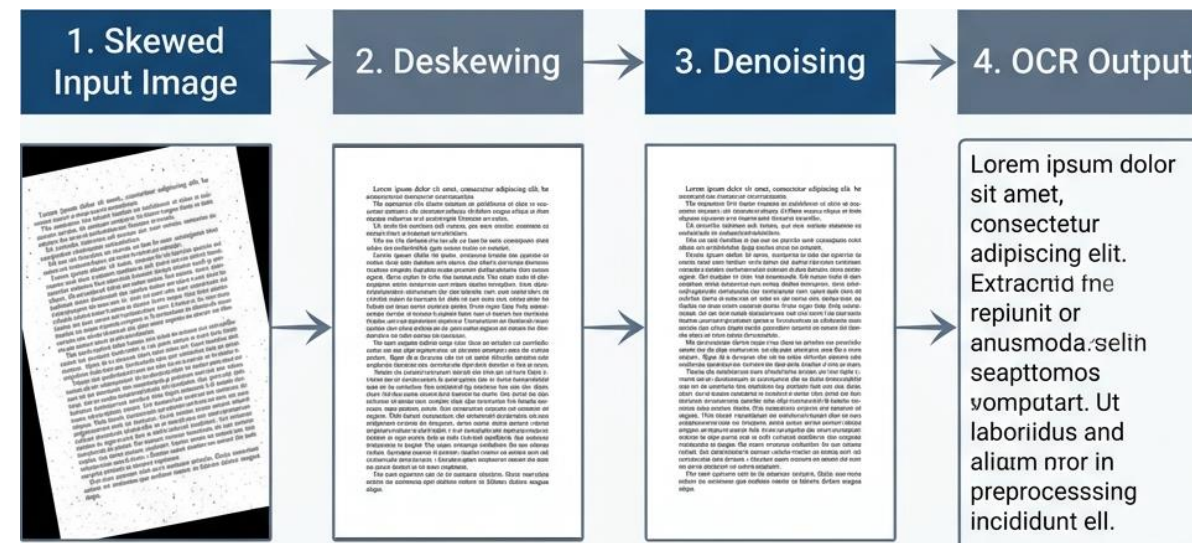


Figure 4.3: OCR Preprocessing Pipeline

Modern Baseline: Vision-Language Models (VLMs)

Multimodal Promising: Treats entire document images natively as input patches.

Resolution Constraints: Fixed patch limits (e.g., 336x336 pixels) heavily compress dense tabular data.

The Perception Failure: Zero-shot VLMs probabilistically "guess" digits when pixel alignment is lost, leading to severe hallucination.

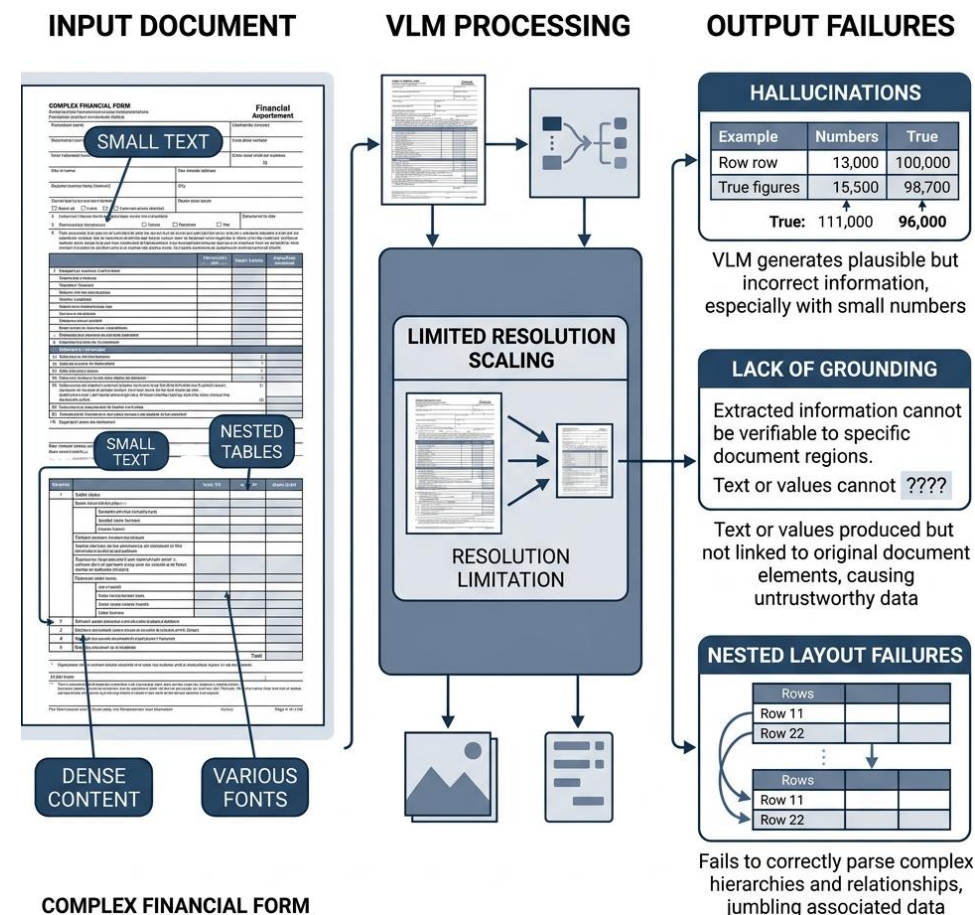


Figure 2.4: VLM Projection Layer and Resolution Constraints

Proposed Solution: Hybrid OCR-VLM

Dual-Stream Synchronization: Merges the literal character precision of PaddleOCR with the semantic layout awareness of VLMs.

Spatial Anchoring: Indexes exact 2D bounding boxes (x_1, y_1, x_2, y_2) for absolute alphanumeric fidelity.

Perception Safety Net: Grounds VLM reasoning inside verified, deterministic OCR coordinates to eliminate hallucination.

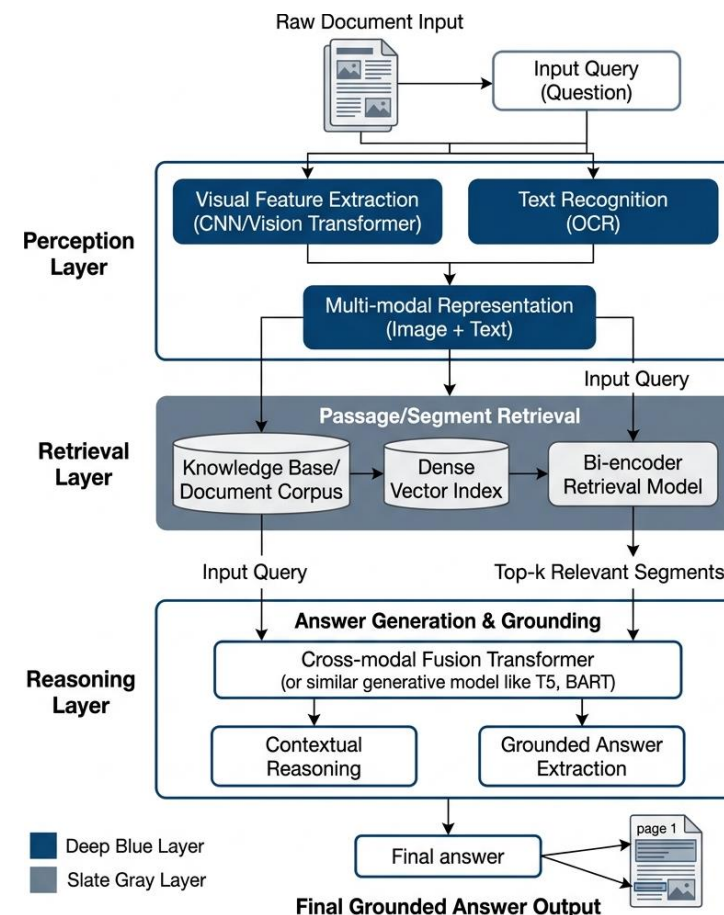


Figure 4.1: Advanced Global System Orchestration Architecture

RAG Connection Layer

Vector Space Mapping: Layout-aware documents are chunked and projected into a FAISS database using `all-MiniLM-L6-v2`.

Hybrid Retrieval: Fetches high-relevance chunks mathematically grounded with precise spatial metadata.

Cognitive Reasoning: Feeds deterministic context directly into the Mistral 7B Instruct reasoning engine.

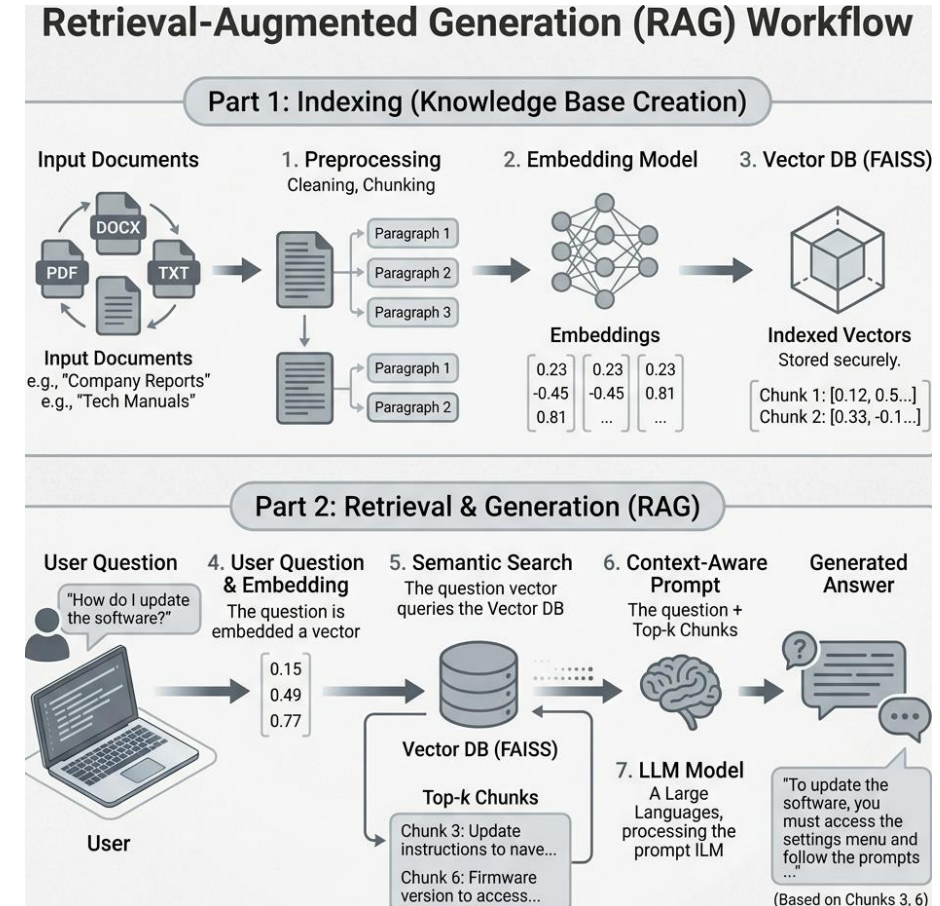


Figure 3.1: Semantic Embedding and Vector Storage Workflow

Evaluation Benchmark & Metrics

Test Corpus: 50 high-complexity zero-shot PDFs from the DocVQA validation set (dense tables, scans, multi-columns).

ANLS (Soft Similarity): Penalizes hallucinations while allowing minor formatting variances.

Exact Match (EM): Strictest binary accuracy.

F1-Score: Token-level lexical overlap measuring precision and recall.

$$ANLS = \frac{1}{N} \sum_{i=1}^N s(a_i, g_i) \quad (3.1)$$

$$EM = \mathbf{1}(a_i = g_i) \quad (3.2)$$

Experimental Results

Empirical Breakthrough: The Hybrid model strictly outperforms standalone VLMs and Traditional OCR.

Visual Evidence: Synchronization successfully prevents layout-blindness.

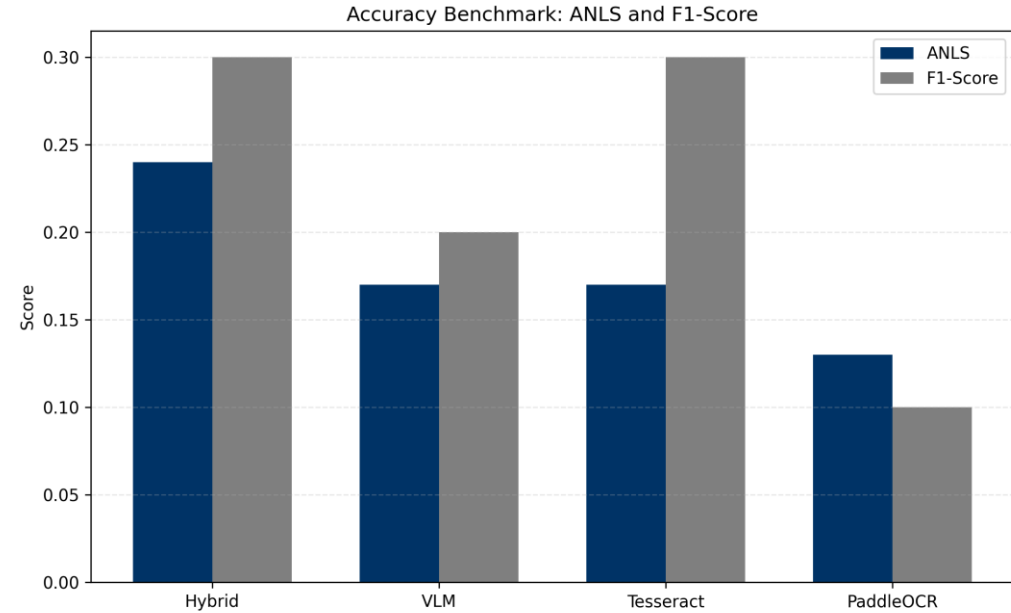


Figure 7.2: Accuracy Benchmark Matrix (ANLS vs F1)

Key Performance Findings

Massive Improvement: The Hybrid model achieves a 41% improvement in ANLS accuracy over standalone VLMs (0.17 $\rightarrow$ 0.24).

Table 7.1: Exhaustive Performance Benchmarking Matrix

System Model	ANLS	Exact Match (EM)	F1-Score	Inference Latency	Throughput	RAM Footprint	Search Efficiency
Hybrid (Ours)	0.24	0.20	0.30	14.20 s	0.07 S/s	4600 MB	Sub-ms (<1ms)
Standalone VLM	0.17	0.10	0.20	4.20 s	0.24 S/s	4100 MB	N/A
Tesseract RAG	0.17	0.10	0.30	11.00 s	0.09 S/s	350 MB	Sub-ms (<1ms)
PaddleOCR RAG	0.13	0.00	0.10	52.30 s	0.02 S/s	850 MB	Sub-ms (<1ms)

Ablation Study: Why it Works

Isolating Variables: We systematically removed the layout parsing and character grounding layers.

Table 7.2: Perception Component Ablation Study
The Core Finding: Removing deterministic character grounding collapses Exact Match accuracy back to zero.

Architecture	Layout Detection	Text Grounding	VLM Context	ANLS	EM
Hybrid (Full)	Yes	Yes	Yes	0.24	0.20
No Grounding	Yes	No	Yes	0.17	0.10
No Layout	No	Yes	Yes	0.13	0.00

Error Analysis & Visual Evidence

VLM Failure Mode: Guessing digits incorrectly due to low patch resolution.

Hybrid Precision: Anchoring logic preserves exact numerical strings.

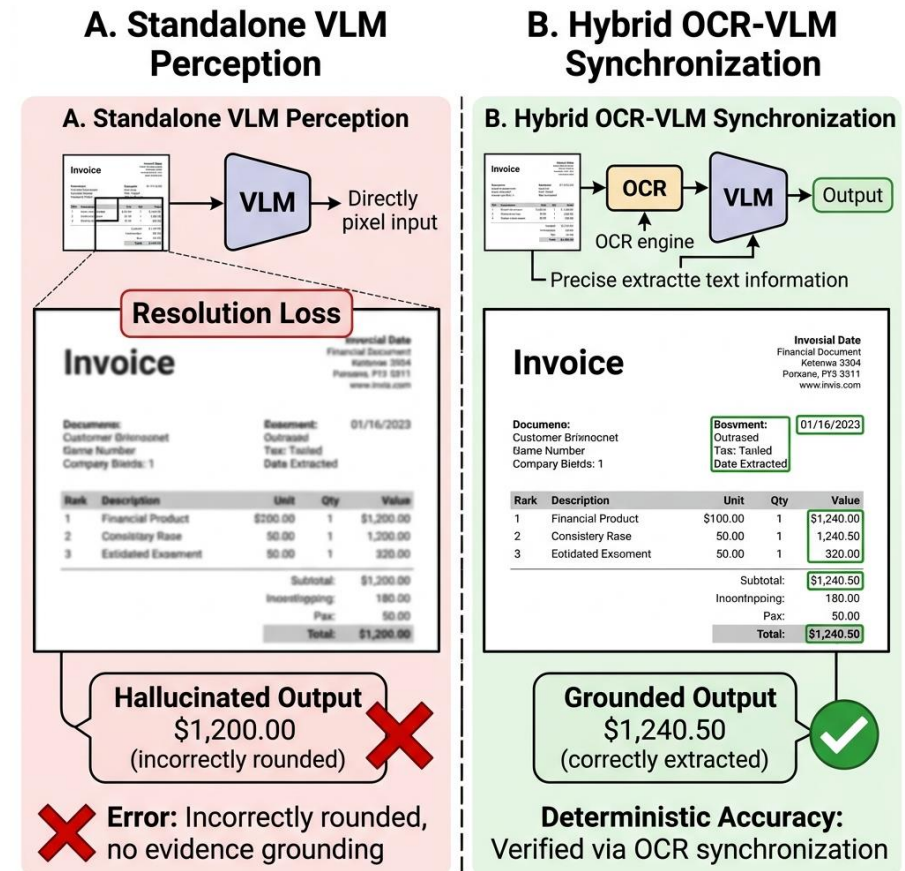


Figure 7.6: Visual Case Study of VLM Resolution-Loss vs. Hybrid Precision

Conclusion

Bridging the Gap: Synchronized OCR-VLM architecture successfully bridges the perception-cognition disconnect.

Grounding Efficacy: Deterministic character grounding is vital for eliminating LLM hallucinations in structured enterprise documents.

Standardized Framework: Provides a repeatable, zero-shot system-level benchmark for future Document AI research.

Future Work: Multi-Agent System

Evolution: Transitioning from a rigid linear pipeline to a cooperative multi-agent network.

Agent Roles:

Layout Agent: Parses boundaries and geometric structures.

OCR Agent: High-fidelity character transcription.

Visual Agent: Processes embedded charts and figures.

Coordinator Agent: Resolves spatial conflicts and synthesizes the final factual answer.

Thank You

Thank You for Your Attention
Questions & Discussion
Tifang Desmond Ngoe
Academic Year 2025-2026